IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
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Practice Problems 03: Generalized Linear Models

Multivariate Calculus Practice

Question 33 (Jacobian computation I). (I) - (II) First find the dimensions, then the Jacobian. It's probably easiest here to use index notation.

a. For the function $f(\mathbf{x}) = \sin(x_1)\cos(x_2)$, find $\frac{\partial f}{\partial \mathbf{x}}$.

b. For the function $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{y}$, find $\frac{\partial f}{\partial \mathbf{x}}$.

c. For the function $f(\mathbf{x}) = \mathbf{x}\mathbf{x}^\top$, find $\frac{\partial f}{\partial \mathbf{x}}$.

d. For the function $f(\mathbf{t}) = \sin(\log(\mathbf{t}^\top \mathbf{t}))$, find $\frac{\partial f}{\partial \mathbf{t}}$.

e. For the function $f(\mathbf{X}) = \text{Tr}(\mathbf{AXB})$ for $\mathbf{A} \in \mathbb{R}^{D\times E}$, $\mathbf{X} \in \mathbb{R}^{E\times F}$, $\mathbf{B} \in \mathbb{R}^{F\times D}$, find $\frac{\partial f}{\partial \mathbf{X}}$.

Question 34 (Chain rule practice I). (II) - (III) Compute the derivatives $\frac{\partial f}{\partial \mathbf{x}}$ of the following functions.

- First, write out the chain rule for the given decomposition.
- Give the shapes of intermediate results, and make clear which dimension(s) will be summed over.
- Provide expressions for the derivatives, and describe your steps in detail. Providing an expression means specifying everything up to the point where you could implement it.
- Give the results in vector notation if you can.

a. For $f(z) = \log(1+z), \quad z = \mathbf{x}^\top \mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^D$.

b. For $f(\mathbf{z}) = \sin(\mathbf{z}), \quad \mathbf{z} = \mathbf{A}\mathbf{x} + \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{E\times D}$. What sizes are $\mathbf{x}$ and $\mathbf{b}$?

c. For $f(z) = \exp\left(-\frac{1}{2}z\right), \quad z = \mathbf{y}^\top \mathbf{S}^{-1}\mathbf{y}, \quad \mathbf{y} = \mathbf{x} - \boldsymbol{\mu}$.

d. For $f(\mathbf{A}) = \text{Tr}(\mathbf{A}), \quad \mathbf{A} = \mathbf{x}\mathbf{x}^\top + \sigma^2\mathbf{I}$.

e. For $\mathbf{f}(\mathbf{z}) = \tanh(\mathbf{z}), \quad \mathbf{z} = \mathbf{A}\mathbf{x} + \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{M \times N}$.

f. For $f(\mathbf{A}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \mathbf{x}\mathbf{x}^\top$.

Question 35 (Hessian Linear Regression). III

For the stationary point of linear regression, find the Hessian, and prove that it is positive definite, perhaps by making some assumptions. Discuss your assumptions.

Topical Applications

Question 36 (Gradient of the Logistic Regression Loss). II

In logistic regression, we model the confidence of class 1 as

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{X}\theta), \qquad \sigma(z) = \frac{1}{1 + e^{-z}},$$

where $\boldsymbol{X} \in \mathbb{R}^{m \times d}$ is the design matrix containing m examples and $\theta \in \mathbb{R}^d$ are the model parameters.

The negative log-likelihood (empirical risk) of this model can be written compactly as

$$\mathcal{L}(\theta) = -\frac{1}{m}\Big[\boldsymbol{y}^\top \log(\sigma(\boldsymbol{X}\theta)) + (\mathbf{1} - \boldsymbol{y})^\top \log(1 - \sigma(\boldsymbol{X}\theta))\Big].$$

Derive the gradient of $\mathcal{L}(\theta)$ with respect to θ.

Question 37 (Bradley–Terry Model and Preference Learning). III

Pairwise comparison models such as the **Bradley–Terry model** are widely used for *preference modeling*—for example, in large language model (LLM) alignment, where human annotators express which of two responses is preferred. Each response (or player) i is assigned a latent *score* parameter β_i, reflecting its underlying quality or skill.

We represent a comparison between two responses i and j using a feature vector $\boldsymbol{x}_{ij} \in \mathbb{R}^n$ defined as:

$$(\boldsymbol{x}_{ij})_k = \begin{cases} +1, & \text{if } k = i, \\ -1, & \text{if } k = j, \\ 0, & \text{otherwise.} \end{cases}$$

Collecting all such feature vectors into a matrix $\boldsymbol{X}$, we also record the outcomes in a label vector $\boldsymbol{y}$, where $y_{ij} = 1$ if i beats j, and 0 otherwise.

(a) Applying a linear model to $\boldsymbol{x}_{ij}$ gives a scalar value $\boldsymbol{x}_{ij}^\top \boldsymbol{\beta}$. Express this term in simpler form, using index notation if helpful. What does it represent intuitively?

(b) To model the probability that i beats j as a classification problem, which nonlinearity from our notes should we apply to $\boldsymbol{x}_{ij}^\top \boldsymbol{\beta}$? Using your answer to part (a), write this probability explicitly.

(c) Derive the gradient of the log-likelihood with respect to $\boldsymbol{\beta}$ for this model.